

- **Inverse Matrix**

An inverse matrix A^{-1} of a square matrix A is a matrix that, when multiplied by A , gives the identity matrix I :

$$AA^{-1} = A^{-1}A = I$$

For a matrix to have an inverse, it must be non-singular (i.e., its determinant must not be zero). The eigenvalues of the inverse matrix A^{-1} are the reciprocals of the eigenvalues of A .

For example, if λ is an eigenvalue of A , then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .

- **Orthonormal Matrix**

An orthonormal matrix (or orthogonal matrix) Q is a square matrix where the rows (and columns) are orthonormal vectors, meaning:

$$QQ^T = Q^TQ = I$$

This implies that the inverse of an orthonormal matrix is equal to its transpose $Q^{-1} = Q^T$. The eigenvalues of an orthonormal matrix have a modulus (absolute value) of 1, meaning they lie on the unit circle in the complex plane.

- **Power Matrix**

The power of a matrix A^n refers to the matrix multiplied by itself n times. The eigenvalues of the matrix power A^n are the eigenvalues of A raised to the n -th power. For example, if λ is an eigenvalue of A , then λ^n is an eigenvalue of A^n .

Problems

a) The eigenvalues of the matrix A are $\{1, 3\}$. Find its A^{-1}

To find the eigenvalues of A^{-1} , take the reciprocal of each eigenvalue of A :

$$\text{For } \lambda_1 = 1, \quad \frac{1}{\lambda_1} = \frac{1}{1} = 1$$

$$\text{For } \lambda_2 = 3, \quad \frac{1}{\lambda_2} = \frac{1}{3}$$

Thus, the eigenvalues of A^{-1} are:

$$\{1, \frac{1}{3}\}$$

b) The eigenvalues of the matrix A are $\{1, 3\}$. Find its A^2

To find the eigenvalues of A^2 , square each eigenvalue of A :

$$\text{For } \lambda_1 = 1, \quad \lambda_1^2 = 1^2 = 1$$

$$\text{For } \lambda_2 = 3, \quad \lambda_2^2 = 3^2 = 9$$

Thus, the eigenvalues of A^2 are:

$$\{1, 9\}$$

Reference: Nicholson, W. K. (2006). Elementary Linear Algebra. ISBN 85-86804-92-4.